Unit 3, Lesson 4: Plotting, Ordering, and Comparing Rational Numbers – Part 2

Benchmark

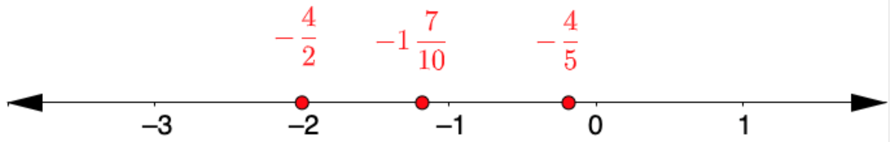
MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

Learning Target

☐ I can plot, compare, and order rational numbers.

3.4.1 Warm-Up: Find the Error

A student plotted the following numbers.



1. Which values are plotted correctly?

2. Which values are NOT plotted correctly?

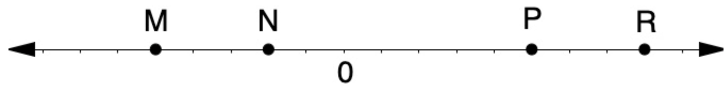
3.4.2 Activity: Class Number Line

Your teacher will give you a sticky note with a rational number written on the front.

Write your name on the back of the sticky note and place your number in the correct location on the class number line.

3.4.3 Exploration: Comparing Points on a Number Line

A number line is shown with points M, N, P, and R. Work with your partner to answer the questions about the number line.



1. Use each of the given phrases to describe or compare the values of points M, N, P, R. Each statement will use one or two of the points.

Phrase	**Description or Comparison of M, N, P or R**
less than	
opposite of	
greater than	
negative number	

2. If P is $2\frac{1}{2}$, what are the values of M, N, and R?

3. If N is -0.4, what are the values of M, P, and R?

4. If R is 200%, what are the values of M, N, and P?

5. If M is $-1\frac{2}{3}$, what are the values of N, P, and R?

3.4.4 Guided Instruction: Comparing Rational Numbers in Different Forms

1. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

9.02 ____ 9.2% $\quad -\frac{3}{5}$ ____ $-\frac{13}{20}$ $\quad 6.050$ ____ 6.05

0.4 ____ $\frac{9}{40}$ $\quad 2\frac{3}{4}$ ____ 2.624 $\quad -4\frac{1}{3}$ ____ $-\frac{10}{3}$

2. Emma and Noah shared their strategy for comparing values written as fractions. Write your strategy in the box and note any **differences** from Emma's or Noah's strategy.

Emma's Strategy	Noah's Strategy
When Emma compares a negative fraction to a negative fraction, she starts by thinking about which whole numbers each fraction lies between and which each is closest to.	When Noah compares a negative fraction to a negative fraction, he uses common denominators.

3. Angel and Sophia share their strategies for comparing a value written as a decimal to a value written as a fraction. Write your strategy in the box and note any **similarities** from Angel's or Sophia's strategy.

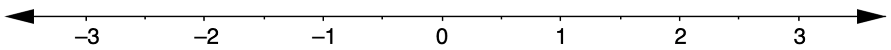
Angel's Strategy	Sophia's Strategy
When Angel compares a positive decimal to a positive fraction, he rewrites the fraction as an equivalent decimal.	When Sophia compares a positive decimal to a positive fraction, she plots them on a number line.

3.4.5 Try It: Plotting, Ordering, and Comparing Rational Numbers in Different Forms

1.

a. Plot each value on the number line. Label each point with its numeric value.

-2.7 $\quad \frac{7}{10}$ $\quad -1\frac{1}{8}$ $\quad 0.4$ $\quad -\frac{5}{2}$ $\quad 28\%$



b. What strategies did you use that were helpful when plotting the rational numbers written in different forms on the number line?

c. Order the values from greatest to least.

d. Complete the inequality statement to show the relationship between the values plotted on the number line.

_______ < _______ < _______ < _______ < _______ < _______

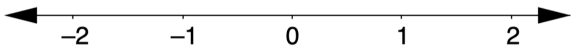
3.4.6 Your Turn!

1. Order the values from least to greatest.

$$\frac{1}{2} \qquad 0 \qquad 1 \qquad -1\frac{1}{2} \qquad -\frac{1}{2} \qquad -1$$

2. Plot the values on the number line. Label each point with its numeric value.

$$0.4 \qquad -1\frac{7}{10} \qquad -\frac{11}{10}$$



3. Order the values from greatest to least.

$$-2 \qquad 0 \qquad -\frac{1}{6} \qquad 1\frac{3}{8} \qquad -3\frac{1}{3} \qquad 1.4 \qquad \frac{5}{4} \qquad -\frac{13}{10} \qquad \frac{21}{5}$$

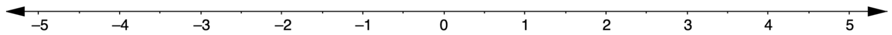
4. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

8.7 ____ 8.689 $\qquad -\frac{5}{3}$ ____ $-\frac{7}{10}$ $\qquad 7.2$ ____ $7\frac{1}{3}$

0.38 ____ 5% $\qquad -4\frac{5}{8}$ ____ -4.5 $\qquad -3\frac{2}{5}$ ____ -3.4

3.4.7 Wrap-Up: Ticket Out the Door

1. Use the values 3.6, $-4\frac{4}{5}$, $0.7, -1\frac{1}{5}$, $\frac{15}{4}$, 89%, and $-\frac{7}{8}$. Plot the values on the number line. Label each point with its numeric value.



2. List the values $3.6, -4\frac{4}{5}, 0.7, -1\frac{1}{5}, \frac{15}{4}, 89\%$, and $-\frac{7}{8}$ from least to greatest.

3. Use the symbols $<$, $>$, or $=$ to compare each pair of values.

-12.742 ____ -12.9 $\qquad 1\frac{2}{9}$ ____ 150% $\qquad 3\frac{7}{12}$ ____ 3.7